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—Devendra Khari, CEO, DKOP

verification (using standard verification methodologies and languages). FPGA designs for a range of market segments are being widely adopted as the system platform for the last few years. The VLSI back-end design includes physical design of the silicon chip and physical verification. The system-/board-level design is a niche area of the training market," explains Wikneswaran Pillai, director, Sandeepani School of VLSI System Design.

Growth drivers

The Indian business climate is conducive to service providers and VLSI and embedded system training is, in fact, a kind of service industry. As a result, this field may be considered as a forte for the Indian players. Pillai explains, "India being a preferred destination for design outsourcing by most companies, training this workforce forms an important part. Also, with India having a huge local market, there are many small and medium companies that have design houses. Graduate pass-outs from engineering colleges are not ready to execute real industry projects and need initial and ongoing training to ensure productivity." Ease of starting operations and affordable training costs for students are some of the other competitive advantages.

According to NASSCOM, the VLSI and embedded industry would employ 750,000 professionals by 2015 with

about 680,000 for embedded systems, 75,000 for VLSI design and the remaining 10,000 in manufacturing. A good number of firms will set up shops in India, which, in turn, will increase the requirement of professionals with the requisite skillsets. The training providers can bridge this gap between the industry requirements and professionals.

The training market grows faster during the economic downturn than in normal circumstances. According to Baker, the reflections of the recent economic slowdown are showing up in the form of cost and infrastructure cuts

"India can address the overseas market as well, but this could be offset to some extent by travel and communications cost. By leveraging the Web and other long-distance methods, these costs can be contained," suggests Khari.

The training institute should also take advantage of the high-tech nature of the field. "In embedded system, the technology changes very fast. Every day each company develops new technologies and products. Embedded engineers trained in the latest technologies are always in demand. Training institutes can play a critical role here," says Nadh.

"For addressing the demand for qualified manpower, a well drawn out and constantly updated training methodology is a must. The stakes placed by the companies in view of our skilled labour and lesser development costs thus are largely dependent

The VLSI training market consists of VLSI front-end design and design verification.

—Wikneswaran Pillai, Director, Sandeepani School of VLSI System Design

among the industries. Big players, for example, are opting for trained professionals against freshers. This scenario is naturally in favour of the training industry. Moreover, even the experienced professionals look upon training institutes to keep abreast of the latest technologies and tools.

Devendra Khari, CEO, DKOP, indicates that 40-50 per cent of people working in this sector need to be re-trained. VLSI/embedded training institutes can address part of this need through corporate training.

on the quality assured by our training centres," explains Baker.

Dr Subbarangaiah feels the need for a win-win interaction amongst semiconductor, product and services companies, government, academia, venture capital providers and industry bodies in developing the ecosystem.

Business strategies

Contrary to general perception of the 'admission→training→placements' model, most industry players prefer to follow unique strategies to ensure sustainability. "We don't limit our operations to training alone, but our R&D and training related activities go hand in hand and the students can benefit a lot from that. They could well achieve a hands-on experience together with our experienced development staff and in a way become aware of the shift in standards from that of a classroom to that of a workbench," says Baker. Everest

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—Dr K. Subbarangaiah, Professor and Director, Veda IIT